LEARNING HUB MASTER SERIES • VOLUME 01

Linear Algebra: Vectors, Linear Combinations & The Column Picture

Foundations of Column Vectors in $\mathbb{R}^n$, Geometric & Algebraic Perspectives, Euclidean Norms, Dot Products, Orthogonal Projections, Cauchy-Schwarz & Triangle Inequalities, Matrix-Vector Multiplication, and Hand-Calculation Fast-Tracks

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Synthesized From:

• Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 1)
• Kenji Hiranabe — *The Art of Linear Algebra* (Visual Column Synthesis & Vector Geometries)
• Learning Hub Interactive Knowledge Base — *Vectors & Linear Combinations* Topics

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Part I: Geometric & Algebraic Foundations of Vectors in $\mathbb{R}^n$

Chapter 1: Three Perspectives on a Vector: Arrows, Arrays, and Elements of $\mathbb{R}^n$

Linear algebra begins with vectors. In conventional education, students often encounter vectors through conflicting definitions depending on whether the instructor is a physicist, a computer scientist, or a pure mathematician. True mastery of linear algebra requires synthesizing all three viewpoints into a unified conceptual framework.

Perspective	Formal Representation	Core Intuition & Invariants
Physics / Geometric	Directed arrow **v** in 2D or 3D space	Characterized entirely by its *magnitude* (length) and *spatial direction*. A geometric vector is translation-invariant: an arrow representing a force of 10 N east is identical regardless of where its tail is rooted in coordinate space. Standard position places the tail at the origin (0, 0, 0).
Computer Science / Data	Ordered tuple or 1D array $[v_1, v_2, \ldots, v_n]^T$	A fixed-dimension column of floating-point numbers where index position encodes distinct features (e.g., patient age, blood pressure, cholesterol). Order is critical: swapping coordinates fundamentally alters the data record.
Mathematician / Pure Algebra	Element of an abstract vector space $\mathbf{v} \in \mathbb{R}^n$	Any algebraic object belonging to a set closed under two foundational operations: **Vector Addition** (**u** + **v**) and **Scalar Multiplication** (**cv**), satisfying the eight formal axioms of vector spaces.

1.1 Column Vectors vs. Row Vectors

In modern matrix algebra (established by Gilbert Strang and standard numerical computing), **vectors are column vectors by default**. A vector in $\mathbb{R}^n$ is represented as an n × 1 matrix:

$$\mathbf{v} = \begin{vmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{vmatrix} \in \mathbb{R}^n, \quad \mathbf{v}^T = \begin{vmatrix} v_1 & v_2 & \cdots & v_n \end{vmatrix} \in \mathbb{R}^{1 \times n}$$

The transpose operator T flips an n × 1 column into a 1 × n row vector. This distinction is central when computing products: the inner product (dot product) is written as row-times-column $\mathbf{u}^T\mathbf{v}$ (yielding a 1 × 1 scalar), whereas the outer product is column-times-row $\mathbf{u}\mathbf{v}^T$ (yielding an n × n rank-one matrix).

Chapter 2: Vector Operations: Addition, Scaling & The Parallelogram Law

The entire machinery of linear algebra rests on exactly two primitive operations performed on column vectors:

1. **Vector Addition:** Combining two vectors **v** and **w** into a single resultant vector **v** + **w** by adding corresponding coordinates:

$$\mathbf{v} + \mathbf{w} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \end{bmatrix}$$

2. **Scalar Multiplication:** Multiplying every entry of **v** by a real scalar $c \in \mathbb{R}$:

$$c\mathbf{v} = c \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} c\,v_1 \\ c\,v_2 \end{bmatrix}$$

2.1 Geometric Interpretation: The Parallelogram and Triangle Laws

Geometrically, vector addition can be visualized through two equivalent constructions:

- **The Triangle Law (Head-to-Tail Chaining):** Place the tail of vector **w** at the tip (arrowhead) of vector **v**. The resultant arrow starting at the tail of **v** and ending at the tip of **w** is **v** + **w**. This demonstrates the associative and path-independent nature of physical displacements.
- **The Parallelogram Law:** Root both vectors **v** and **w** at the origin (0,0). Complete the parallelogram whose adjacent sides are **v** and **w**. The main diagonal extending from the origin to the opposite vertex represents the sum **v** + **w**. The other diagonal, connecting the tip of **w** to the tip of **v**, represents the difference vector **v** − **w**.

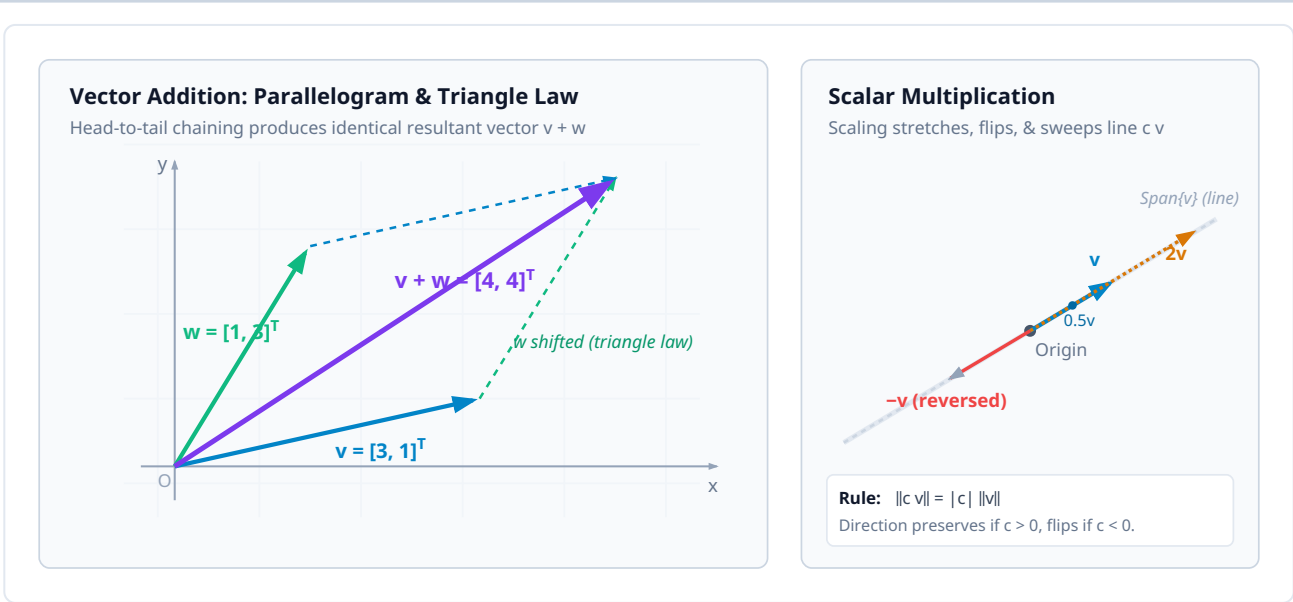


Figure 1.1: *Geometric construction of vector addition in $\mathbb{R}^2$. Left: The Parallelogram Law and Triangle Law demonstrating that chaining* **v** *= [3, 1]^T and* **w** *= [1, 3]^T produces* **v** + **w** *= [4, 4]^T. Right: Scalar multiplication stretching, flipping, and sweeping out the 1D subspace line Span{**v**}.*

⚡ Fast-Track 1: Mental 2D Perpendicularity via Swap-and-Negate

To instantly construct a vector perpendicular to any 2D vector $\mathbf{v} = [a, b]^T$ without solving equations:

$$\mathbf{v}^{\perp} = \begin{vmatrix} -b \\ a \end{vmatrix} \quad \text{or} \quad \begin{vmatrix} b \\ -a \end{vmatrix}$$

Verification: $[a, b] \cdot [-b, a]^T = -ab + ba = 0$. Swapping entries and negating exactly one coordinate rotates the vector by 90° counterclockwise (for $[-b, a]^T$) or 90° clockwise (for $[b, -a]^T$). In higher dimensions, zero-sum pairs allow instantaneous mental verification of orthogonality.

Part II: Geometric Span & Linear Combinations

Chapter 3: Linear Combinations: Filling Lines, Planes, and 3D Spaces

The central operation of linear algebra is taking a set of vectors, scaling each by an independent scalar weight, and summing them together. This composite operation is known as a **linear combination**:

Definition: Linear Combination

Given vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k \in \mathbb{R}^n$ and real scalars $c_1, c_2, \ldots, c_k \in \mathbb{R}$, the vector

$$\mathbf{w} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_k\mathbf{v}_k = \sum_{i=1}^{k} c_i \mathbf{v}_i$$

is called a **linear combination** of $\mathbf{v}_1, \ldots, \mathbf{v}_k$. The collection of *all possible* linear combinations formed by allowing the scalars c_i to range across all real numbers $\mathbb{R}$ is the **Span** of the vectors:

$$\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k\} = \{ \sum_{i=1}^{k} c_i \mathbf{v}_i \quad | \quad c_i \in \mathbb{R} \}$$

3.1 Geometric Hierarchy of Combinations in 3D Space ($\mathbb{R}^3$)

Gilbert Strang emphasizes the geometric progression that occurs as vectors are added to a combination:

- **Zero Vectors (The Origin):** With no vectors (or only the zero vector $\mathbf{0}$), the only combination is the single isolated point (0, 0, 0), a zero-dimensional subspace.
- **One Non-Zero Vector $c\mathbf{v}$ (A Line):** All scalar multiples $c\mathbf{v}$ sweep out an infinite straight line passing through the origin in the direction of $\mathbf{v}$. Dimension = 1.
- **Two Non-Collinear Vectors $c\mathbf{v} + d\mathbf{w}$ (A Plane):** If $\mathbf{w}$ is not on the line of $\mathbf{v}$ (meaning $\mathbf{w} \neq k\mathbf{v}$), all linear combinations $c\mathbf{v} + d\mathbf{w}$ sweep out a completely flat 2D plane passing through the origin. Dimension = 2.
- **Three Non-Coplanar Vectors $c\mathbf{u} + d\mathbf{v} + e\mathbf{w}$ (All of $\mathbb{R}^3$):** If the third vector $\mathbf{u}$ does not lie in the plane spanned by $\mathbf{v}$ and $\mathbf{w}$, their combinations fill the entire 3D space $\mathbb{R}^3$. Dimension = 3.

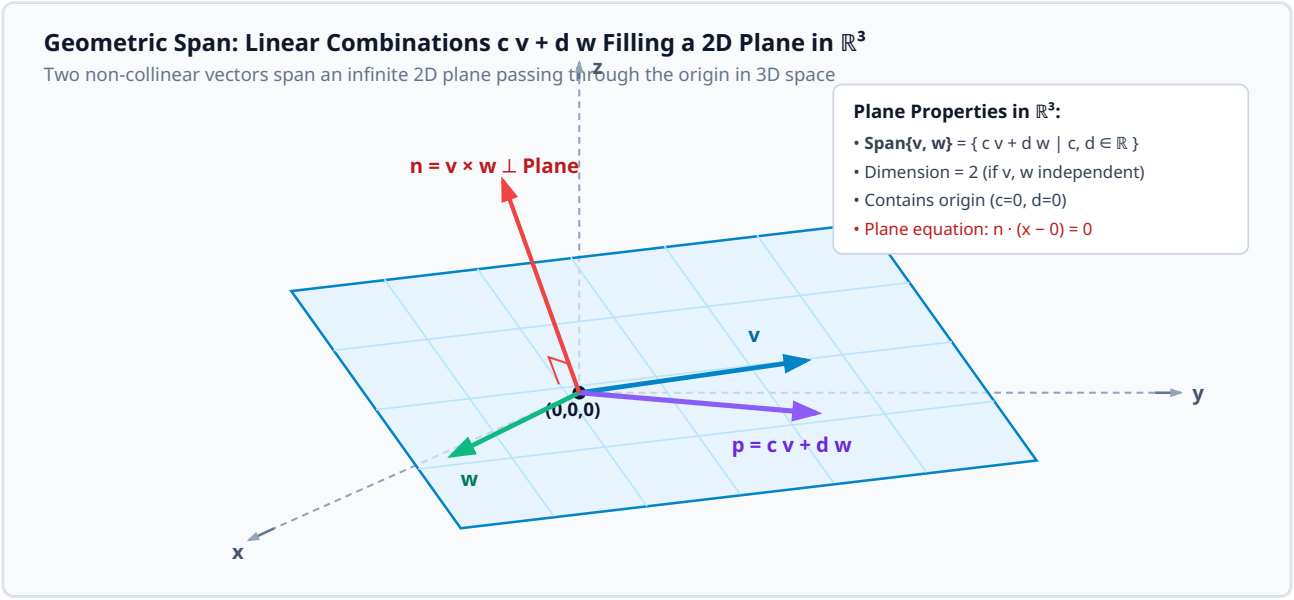


Figure 2.1: *Geometric representation of the linear span Span{**v**, **w**} filling an infinite 2D plane passing through the origin in 3D Euclidean space $\mathbb{R}^3$. Every point **p** on the grid is uniquely addressed by its coordinates (c, d) along the basis arrows. The normal vector **n** = **v** × **w** defines the orthogonal direction perpendicular to the entire plane.*

Chapter 4: Specialized Families: Affine Subspaces, Convex Hulls & Standard Basis Sets

By placing algebraic constraints on the scalar coefficients c_i, linear combinations specialize into powerful geometric objects widely used in computer graphics, optimization, and game theory:

Combination Type	Algebraic Constraint	Geometric Structure (2 Vectors)	Geometric Structure (3 Vectors)
Linear Combination	$c_i \in \mathbb{R}$ (Unconstrained)	Infinite 2D plane through origin	Infinite 3D space $\mathbb{R}^3$ (if independent)
Affine Combination	$\sum c_i = 1$	Infinite straight line through heads of **v**, **w**	Infinite 2D plane passing through tips of **u**, **v**, **w** (not origin)
Convex Combination	$\sum c_i = 1$ and $c_i \geq 0$	Closed line segment joining **v** and **w**	Filled solid triangle spanned by **u**, **v**, **w**
Conic Combination	$c_i \geq 0$ (Non-negative)	Infinite 2D planar wedge (cone)	Infinite 3D convex polyhedral cone

The Standard Basis of $\mathbb{R}^n$

The simplest and most fundamental linear combinations are those using the standard Cartesian basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, …, $\mathbf{e}_n$:

$$\mathbf{e}_1 = \begin{vmatrix} 1 \\ 0 \\ 0 \end{vmatrix}, \quad \mathbf{e}_2 = \begin{vmatrix} 0 \\ 1 \\ 0 \end{vmatrix}, \quad \mathbf{e}_3 = \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} \quad \&implies; \quad \mathbf{v} = \begin{vmatrix} v_1 \\ v_2 \\ v_3 \end{vmatrix} = v_1\mathbf{e}_1 + v_2\mathbf{e}_2 + v_3\mathbf{e}_3$$

Every column vector in $\mathbb{R}^n$ is automatically a unique linear combination of the standard basis vectors, where the components v_i serve directly as the combination weights.

Part III: Lengths, Dot Products & Euclidean Geometry

Chapter 5: The Dot Product, Euclidean Norms & Unit Vector Normalization

While vector addition and scalar multiplication define the algebraic skeleton of a vector space, they contain no concept of distance, angle, or perpendicularity. The **dot product** (or Euclidean inner product) provides the metric engine that injects geometry into algebra.

Definition: The Algebraic Dot Product

For two column vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the dot product is defined as the sum of products of matching entries, written in matrix notation as row-times-column:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = \begin{vmatrix} u_1 & u_2 & \cdots & u_n \end{vmatrix} \begin{vmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{vmatrix} = u_1v_1 + u_2v_2 + \cdots + u_nv_n = \sum_{i=1}^{n} u_i v_i$$

5.1 Fundamental Algebraic Properties of the Dot Product

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and real scalars $c \in \mathbb{R}$, the dot product satisfies four universal axioms:

1. **Commutativity (Symmetry):** $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$.
2. **Distributivity over Addition:** $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$.
3. **Bilinearity / Scalar Factoring:** $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$.
4. **Positive Definiteness:** $\mathbf{v} \cdot \mathbf{v} \geq 0$ for all $\mathbf{v}$, and $\mathbf{v} \cdot \mathbf{v} = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

5.2 The Euclidean Norm (Length)

The dot product of a vector with itself is the sum of squares of its components. Taking the square root defines the standard **Euclidean Norm** (L2 norm or geometric magnitude) $\|\mathbf{v}\|$:

$$\|\mathbf{v}\| = \sqrt{(\mathbf{v} \cdot \mathbf{v})} = \sqrt{(\mathbf{v}^T\mathbf{v})} = \sqrt{(v_1^2 + v_2^2 + \cdots + v_n^2)}$$

Properties of the Euclidean norm:

- **Non-negativity:** $\|\mathbf{v}\| \geq 0$, with $\|\mathbf{v}\| = 0 \Leftrightarrow \mathbf{v} = \mathbf{0}$.
- **Absolute Homogeneity:** $\|c\mathbf{v}\| = |c| \, \|\mathbf{v}\|$ (scaling scales length by absolute value).
- **Subadditivity:** $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ (The Triangle Inequality).

5.3 Unit Vectors & Direction Cosines

A vector $\mathbf{u}$ is called a **unit vector** if its Euclidean norm equals 1: $\|\mathbf{u}\| = 1$. Any non-zero vector $\mathbf{v} \neq \mathbf{0}$ is normalized into a pure unit direction vector by dividing by its length:

$$\mathbf{u} = \frac{\mathbf{v}}{\|\mathbf{v}\|} \quad \&implies; \quad \|\mathbf{u}\| = \frac{\|\mathbf{v}\|}{\|\mathbf{v}\|} = 1$$

In $\mathbb{R}^2$, every unit vector lies on the unit circle: $\mathbf{u} = [\cos\theta, \sin\theta]^T$. In $\mathbb{R}^3$, a unit vector encodes the **direction cosines** relative to the standard x, y, z axes: $\mathbf{u} = [\cos\alpha, \cos\beta, \cos\gamma]^T$, satisfying the identity $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

⚡ Fast-Track 2: Unit Vector Normalization & Pythagorean Integer Sets

Recognizing integer norm triples and quadruples eliminates square roots during hand calculations:

- **2D Integer Pairs:** [3, 4] → $\|v\| = 5$; [5, 12] → $\|v\| = 13$; [8, 15] → $\|v\| = 17$; [7, 24] → $\|v\| = 25$.
- **3D Integer Triples:** [1, 2, 2] → $\|v\| = \sqrt{(1+4+4)} = \mathbf{3}$; [2, 3, 6] → $\sqrt{(4+9+36)} = \mathbf{7}$; [1, 4, 8] → $\sqrt{(1+16+64)} = \mathbf{9}$; [4, 4, 7] → $\sqrt{(16+16+49)} = \mathbf{9}$; [2, 6, 9] → $\sqrt{(4+36+81)} = \mathbf{11}$.
- **Scaling Shortcut:** If computing the norm of $[12, -16, 24]^T$, factor out gcd=4 first: $4 \times \|[3, -4, 6]^T\| = 4\sqrt{(9+16+36)} = 4\sqrt{61}$. Never square large numbers directly!

Chapter 6: Angles, Orthogonality, Law of Cosines & Orthogonal Projections

6.1 The Geometric Dot Product & Law of Cosines Derivation

The dot product establishes the direct trigonometric link between coordinate algebra and spatial angles:

Theorem: Geometric Formula for the Dot Product

For any two non-zero vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ forming an interior angle $\theta \in [0, \pi]$:

$$\mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\| \, \|\mathbf{w}\| \cos\theta \quad \Leftrightarrow \quad \cos\theta = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \, \|\mathbf{w}\|}$$

Rigorous Derivation via the Classical Law of Cosines

Consider the triangle in $\mathbb{R}^n$ with side vectors $\mathbf{v}$, $\mathbf{w}$, and the closing side $\mathbf{v} - \mathbf{w}$. By the classical geometric Law of Cosines:

$$\|\mathbf{v} - \mathbf{w}\|^2 = \|\mathbf{v}\|^2 + \|\mathbf{w}\|^2 - 2\|\mathbf{v}\|\|\mathbf{w}\| \cos\theta$$

Now expand the algebraic dot product of the LHS:

$$\|\mathbf{v} - \mathbf{w}\|^2 = (\mathbf{v} - \mathbf{w}) \cdot (\mathbf{v} - \mathbf{w}) = \mathbf{v}\cdot\mathbf{v} - 2(\mathbf{v}\cdot\mathbf{w}) + \mathbf{w}\cdot\mathbf{w} = \|\mathbf{v}\|^2 - 2(\mathbf{v}\cdot\mathbf{w}) + \|\mathbf{w}\|^2$$

Equating the two expressions and subtracting $\|\mathbf{v}\|^2 + \|\mathbf{w}\|^2$ from both sides:

$$-2(\mathbf{v} \cdot \mathbf{w}) = -2\|\mathbf{v}\|\|\mathbf{w}\| \cos\theta \quad \&implies; \quad \mathbf{v} \cdot \mathbf{w} = \|\mathbf{v}\|\|\mathbf{w}\| \cos\theta. \quad \square$$

6.2 Orthogonality (Perpendicularity)

When the angle between two vectors is θ = 90° (π/2 radians), cosθ = 0. This produces the single most important definition in linear algebra:

Definition: Orthogonal Vectors

Two vectors **v** and **w** are **orthogonal** (written **v** ⊥ **w**) if and only if their dot product vanishes:

$$\mathbf{v} \perp \mathbf{w} \quad \Leftrightarrow \quad \mathbf{v} \cdot \mathbf{w} = 0$$

Significance of the Dot Product Sign:

- $\mathbf{v} \cdot \mathbf{w} > 0 \Leftrightarrow 0° \le \theta < 90°$ (Acute angle; vectors generally agree in direction).
- $\mathbf{v} \cdot \mathbf{w} = 0 \Leftrightarrow \theta = 90°$ (Right angle; completely independent orthogonal directions).
- $\mathbf{v} \cdot \mathbf{w} < 0 \Leftrightarrow 90° < \theta \le 180°$ (Obtuse angle; vectors oppose each other).

6.3 Orthogonal Projection onto a Line

Projecting a vector **v** onto the direction of a non-zero vector **w** breaks **v** into two perpendicular components: a component **p** parallel to **w**, and an error component **e** strictly perpendicular to **w**:

$$\mathbf{v} = \mathbf{p} + \mathbf{e}, \quad \text{where} \quad \mathbf{p} = c\mathbf{w} \quad \text{and} \quad \mathbf{e} \perp \mathbf{w}$$

To determine the unknown scalar coefficient c, enforce the orthogonality condition $\mathbf{e} \cdot \mathbf{w} = 0$:

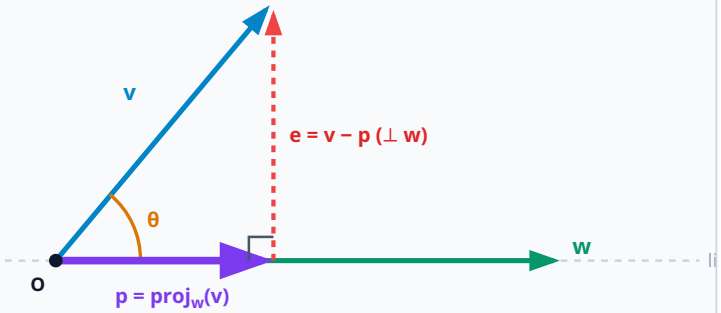
$$(\mathbf{v} - c\mathbf{w}) \cdot \mathbf{w} = 0 \quad \&implies; \quad \mathbf{v} \cdot \mathbf{w} - c(\mathbf{w} \cdot \mathbf{w}) = 0 \quad \&implies; \quad c = \frac{\mathbf{v} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}} = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{w}\|^2}$$

The orthogonal projection vector **p** is therefore:

$$\mathbf{p} = \text{proj}_{\mathbf{w}}(\mathbf{v}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\mathbf{w} \cdot \mathbf{w}} \mathbf{w} = (\mathbf{v} \cdot \hat{\mathbf{u}})\, \hat{\mathbf{u}}, \quad \text{where} \quad \hat{\mathbf{u}} = \mathbf{w} / \|\mathbf{w}\|$$

Dot Product Geometry & Orthogonal Projection

Inner product measures alignment: v · w = ‖v‖ ‖w‖ cosθ, decomposing v into parallel p and perpendicular e



Key Projection Formulas

1. Scalar Projection (Component):

$\text{comp}_w(v) = \|v\| \cos\theta = (v \cdot w) / \|w\|$

2. Orthogonal Projection Vector:

$p = [\,(v \cdot w) / (w \cdot w)\,]\, w$

3. Orthogonality Condition:

$e \cdot w = (v - p) \cdot w = 0$

4. Pythagorean Decomposition:

$\|v\|^2 = \|p\|^2 + \|e\|^2$

$\Rightarrow \|p\| \le \|v\|$ (Cauchy-Schwarz)

Figure 3.1: *Geometric decomposition of vector* **v** *into its orthogonal projection* $\mathbf{p} = \text{proj}_{\mathbf{w}}(\mathbf{v})$ *along* **w** *and perpendicular error vector* $\mathbf{e} = \mathbf{v} - \mathbf{p}$. *Right-angle marker confirms* $\mathbf{e} \perp \mathbf{w}$, *forming a right triangle with* $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{e}\|^2$.

Part IV: Fundamental Geometric Inequalities & Rigorous Proofs

Chapter 7: The Cauchy-Schwarz Inequality: Theorem, Discriminant Proof & Equality Conditions

The Cauchy-Schwarz inequality is hailed by mathematicians as one of the most powerful and fundamental inequalities in all of mathematical analysis. It places an absolute upper bound on the dot product of two vectors in terms of their Euclidean lengths.

Theorem: The Cauchy-Schwarz Inequality

For any two vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$| \mathbf{v} \cdot \mathbf{w} | \leq \|\mathbf{v}\| \, \|\mathbf{w}\|$$

Written out in component coordinates:

$$\left| \sum_{i=1}^{n} v_i w_i \right| \leq \sqrt{\left(\sum_{i=1}^{n} v_i^2 \right)} \cdot \sqrt{\left(\sum_{i=1}^{n} w_i^2 \right)}$$

Equality Condition: Equality $|\mathbf{v} \cdot \mathbf{w}| = \|\mathbf{v}\| \, \|\mathbf{w}\|$ holds if and only if $\mathbf{v}$ and $\mathbf{w}$ are linearly dependent (collinear), meaning one is a scalar multiple of the other: $\mathbf{w} = k\mathbf{v}$ for some $k \in \mathbb{R}$.

Rigorous Algebraic Proof via Quadratic Discriminant

Step 1: Trivial Case. If $\mathbf{v} = \mathbf{0}$ or $\mathbf{w} = \mathbf{0}$, both sides evaluate to 0, and equality trivially holds. Assume $\mathbf{v} \neq \mathbf{0}$.

Step 2: Construct a Non-Negative Quadratic Function. For any real variable $t \in \mathbb{R}$, consider the squared Euclidean norm of the vector linear combination $t\mathbf{v} + \mathbf{w}$:

$$f(t) = \| t\mathbf{v} + \mathbf{w} \|^2 \geq 0 \qquad (\forall\, t \in \mathbb{R})$$

Step 3: Expand the Dot Product. Using the algebraic bilinearity and symmetry of the dot product:

$$f(t) = (t\mathbf{v} + \mathbf{w}) \cdot (t\mathbf{v} + \mathbf{w}) = t^2\,(\mathbf{v} \cdot \mathbf{v}) + 2t\,(\mathbf{v} \cdot \mathbf{w}) + (\mathbf{w} \cdot \mathbf{w})$$

Notice that this is a classic quadratic polynomial in t of the standard form $At^2 + Bt + C$, where:

$$A = \|\mathbf{v}\|^2 > 0, \qquad B = 2(\mathbf{v} \cdot \mathbf{w}), \qquad C = \|\mathbf{w}\|^2$$

Step 4: Analyze the Discriminant. Since $f(t) \geq 0$ for *all* real t, the parabolic curve $y = f(t)$ opens upward and never dips strictly below the horizontal t-axis. Therefore, $f(t) = 0$ can have at most **one real root** (a single tangent touching point). This strictly requires the quadratic discriminant Δ to be less than or equal to zero:

$$\Delta = B^2 - 4AC \leq 0$$

Substituting the vector values into Δ:

$$[2(\mathbf{v} \cdot \mathbf{w})]^2 - 4(\|\mathbf{v}\|^2)(\|\mathbf{w}\|^2) \leq 0 \quad \&implies; \quad 4(\mathbf{v} \cdot \mathbf{w})^2 \leq 4\|\mathbf{v}\|^2\,\|\mathbf{w}\|^2$$

Dividing both sides by 4:

$$(\mathbf{v} \cdot \mathbf{w})^2 \leq \|\mathbf{v}\|^2\,\|\mathbf{w}\|^2$$

Taking the non-negative square root of both sides immediately yields:

$$|\, \mathbf{v} \cdot \mathbf{w} \,| \leq \|\mathbf{v}\|\,\|\mathbf{w}\|. \qquad \square$$

Step 5: Equality Analysis. Equality $\Delta = 0$ occurs if and only if there exists a scalar t_0 such that $\|t_0\mathbf{v} + \mathbf{w}\|^2 = 0$. By positive definiteness, this requires $t_0\mathbf{v} + \mathbf{w} = \mathbf{0}$ &implies; $\mathbf{w} = -t_0\mathbf{v}$. Thus, equality holds precisely when the vectors are collinear.

⚡ Fast-Track 3: Cauchy-Schwarz Equality Inspection & Rational Angle Test

Collinearity Inspection: Before calculating norms, check if coordinate ratios are constant:

$$\frac{v_1}{w_1} = \frac{v_2}{w_2} = \cdots = \frac{v_n}{w_n} = k$$

If all ratios match, $|\mathbf{v} \cdot \mathbf{w}| = \|\mathbf{v}\| \, \|\mathbf{w}\|$ automatically (0% calculation required). If even one coordinate ratio differs, the strict inequality $|\mathbf{v} \cdot \mathbf{w}| < \|\mathbf{v}\| \, \|\mathbf{w}\|$ holds, guaranteeing that the angle θ is strictly between 0° and 180°.

Chapter 8: The Triangle Inequality, Reverse Form & Euclidean Metric Distances

The Triangle Inequality states that the shortest path between two points in Euclidean space is a direct straight line: the third side of a triangle cannot exceed the sum of the other two sides.

Theorem: The Triangle Inequality

For any vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$\| \mathbf{v} + \mathbf{w} \| \leq \| \mathbf{v} \| + \| \mathbf{w} \|$$

Equality Condition: Equality $\|\mathbf{v} + \mathbf{w}\| = \|\mathbf{v}\| + \|\mathbf{w}\|$ holds if and only if one vector is a non-negative scalar multiple of the other ($\mathbf{w} = c\mathbf{v}$ with $c \geq 0$), meaning both vectors point in the identical spatial direction.

Rigorous Derivation via Cauchy-Schwarz

Expand the squared norm of the sum $\mathbf{v} + \mathbf{w}$:

$$\| \mathbf{v} + \mathbf{w} \|^2 = (\mathbf{v} + \mathbf{w}) \cdot (\mathbf{v} + \mathbf{w}) = \|\mathbf{v}\|^2 + 2(\mathbf{v} \cdot \mathbf{w}) + \|\mathbf{w}\|^2$$

Applying the property that any real number is less than or equal to its absolute value, followed by the Cauchy-Schwarz inequality $\mathbf{v} \cdot \mathbf{w} \leq |\mathbf{v} \cdot \mathbf{w}| \leq \|\mathbf{v}\|\|\mathbf{w}\|$:

$$\| \mathbf{v} + \mathbf{w} \|^2 \leq \|\mathbf{v}\|^2 + 2 \, |\mathbf{v} \cdot \mathbf{w}| + \|\mathbf{w}\|^2 \leq \|\mathbf{v}\|^2 + 2 \, \|\mathbf{v}\| \, \|\mathbf{w}\| + \|\mathbf{w}\|^2$$

Notice that the RHS is a perfect square:

$$\| \mathbf{v} + \mathbf{w} \|^2 \leq (\|\mathbf{v}\| + \|\mathbf{w}\|)^2$$

Taking square roots of both non-negative terms yields:

$$\| \mathbf{v} + \mathbf{w} \| \leq \| \mathbf{v} \| + \| \mathbf{w} \|. \quad \square$$

8.1 The Reverse Triangle Inequality

A direct corollary provides a lower bound on the length of the vector difference:

$$| \|\mathbf{v}\| - \|\mathbf{w}\| | \leq \| \mathbf{v} - \mathbf{w} \|$$

Together, the forward and reverse triangle inequalities establish the **Euclidean Metric Space** distance axioms for $d(\mathbf{v}, \mathbf{w}) = \|\mathbf{v} - \mathbf{w}\|$:

1. $d(\mathbf{v}, \mathbf{w}) \geq 0$, with $d(\mathbf{v}, \mathbf{w}) = 0 \Leftrightarrow \mathbf{v} = \mathbf{w}$.
2. $d(\mathbf{v}, \mathbf{w}) = d(\mathbf{w}, \mathbf{v})$ (Symmetry).
3. $d(\mathbf{u}, \mathbf{w}) \leq d(\mathbf{u}, \mathbf{v}) + d(\mathbf{v}, \mathbf{w})$ (Metric Triangle Inequality).

Part V: The Column Picture of Matrix-Vector Systems

Chapter 9: Strang's Fundamental Column Picture: Ax as a Combination of Columns

Most elementary textbooks teach matrix multiplication through the cumbersome row-by-column dot product rule. While algebraically valid, this "row picture" completely conceals the geometric soul of linear algebra. Gilbert Strang's foundational insight is that **matrix-vector multiplication is literally a linear combination of the columns of the matrix**.

The Fundamental Column Law of Matrix Multiplication

Let A be an $m \times n$ matrix whose n columns are vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n \in \mathbb{R}^m$, and let $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$ be a column vector of scalar weights. Then:

$$A\mathbf{x} = \begin{bmatrix} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \sum_{j=1}^{n} x_j \mathbf{a}_j$$

This single equation transforms linear algebra:

- The numbers inside $\mathbf{x}$ are not just coordinates; they are **multipliers (weights)** scaling the column arrows of matrix A.
- The output vector $\mathbf{b} = A\mathbf{x}$ is synthesized by stretching each column vector by its respective weight and chaining them together head-to-tail.
- The set of all possible outputs $A\mathbf{x}$ as $\mathbf{x}$ varies over all of $\mathbb{R}^n$ is the **Column Space** $C(A) = \text{Span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$.

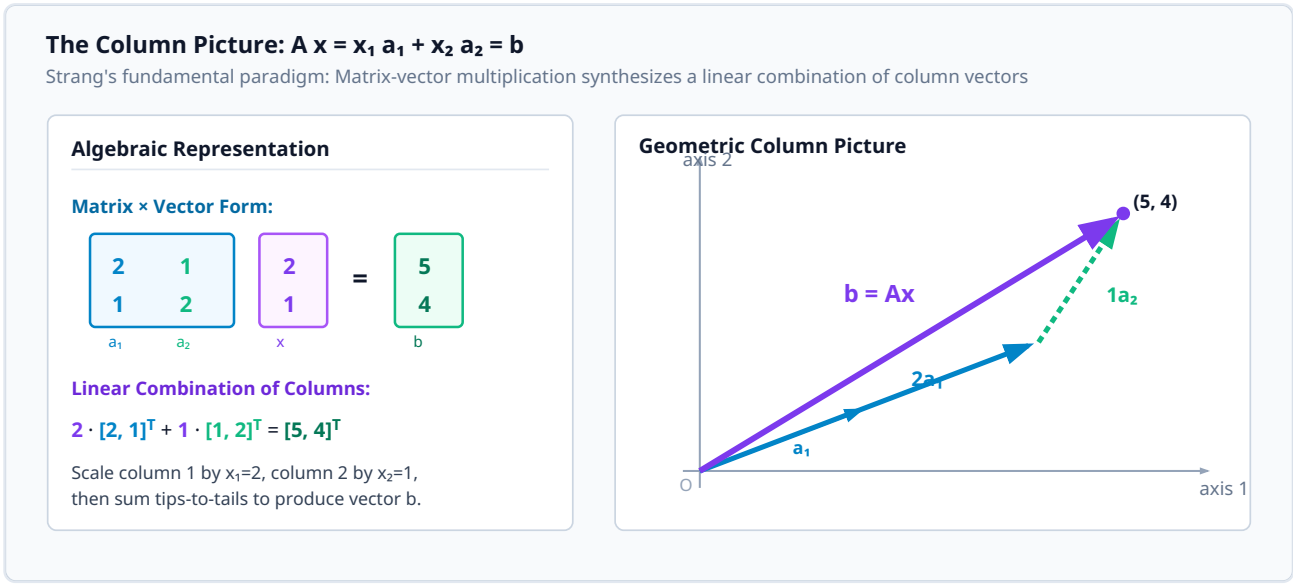


Figure 4.1: *The Column Picture of A**x** = **b**. Left: Matrix-vector equation partitioned into columns scaled by scalar entries $x_1 = 2$ and $x_2 = 1$. Right: Geometric synthesis in the plane, showing $2\mathbf{a}_1 + 1\mathbf{a}_2$ chaining head-to-tail to reach the target vector **b** = [5, 4]^T.*

Chapter 10: Solving Ax = b: Column Synthesis vs. Row Intersection

When solving a system of linear equations A**x** = **b**, two radically different geometric perspectives emerge:

Feature	The Row Picture	The Column Picture (Strang's View)
Mathematical View	Each equation represents a hyperplane: $(\text{row}_i) \cdot \mathbf{x} = b_i$	Single vector synthesis: $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$
Geometric Action	Finding the single point of mutual intersection among m lines/planes	Finding the correct scalar weights to combine n column vectors to reach **b**
Solvability Criterion	Intersections exist without parallel conflicting planes	Target vector **b** lies inside the column space: $\mathbf{b} \in C(A)$
Dimensionality	Visualized in the n-dimensional space of unknowns (x, y, z)	Visualized in the m-dimensional space of data / outputs

⚡ Fast-Track 4: The Column Combination Speed-Check for Ax = b

Before writing down an augmented matrix for Gaussian elimination, inspect the columns of A against **b**:

- **Single Column Match:** If **b** equals column j of A, then $\mathbf{x} = \mathbf{e}_j$ (the j-th standard unit vector). E.g., if $\mathbf{b} = \mathbf{a}_2$, then $\mathbf{x} = [0, 1, 0]^T$.
- **Sum of Columns:** If **b** equals the sum of all columns of A, then $\mathbf{x} = [1, 1, \ldots, 1]^T$.
- **Column Difference:** If $\mathbf{b} = \mathbf{a}_1 - \mathbf{a}_2$, then $\mathbf{x} = [1, -1, 0, \ldots]^T$.
- **Zero Vector:** A**x** = **0** always has the trivial solution **x** = **0**. Non-zero solutions exist if and only if the columns are linearly dependent.

Chapter 11: Difference Matrices, Cyclic Differences & Linear Independence

Gilbert Strang introduces linear algebra through two magnificent classes of matrices that illustrate difference equations (the discrete analogs of derivatives):

11.1 The Forward / Backward Difference Matrix

Consider the 3 × 3 first-difference matrix A:

$$A = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} \quad \&implies; \quad A\mathbf{x} = \begin{vmatrix} x_1 \\ x_2 - x_1 \\ x_3 - x_2 \end{vmatrix} = \begin{vmatrix} b_1 \\ b_2 \\ b_3 \end{vmatrix}$$

Solving A**x** = **b** by back-substitution reveals that **x** computes the **cumulative running sum** of **b**:

$$x_1 = b_1, \quad x_2 = b_1 + b_2, \quad x_3 = b_1 + b_2 + b_3$$

Thus, the inverse of the difference matrix A is the lower-triangular **Sum Matrix S**:

$$S = A^{-1} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix}$$

This mirrors calculus: **Differentiation (differences A) is inverted by Integration (sums S)**. The columns of A are linearly independent, and its column space is all of $\mathbb{R}^3$.

11.2 The Cyclic Difference Matrix (Singularity & Dependence)

Now introduce cyclic boundary conditions where the final row differences back to the first variable:

$$C = \begin{vmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix} \quad \&implies; \quad C\mathbf{x} = \begin{vmatrix} x_1 - x_3 \\ x_2 - x_1 \\ x_3 - x_2 \end{vmatrix}$$

Singularity of the Cyclic Difference Matrix

Notice that the sum of the rows of C is [0, 0, 0]. Likewise, the sum of all three columns is **0**:

$$\mathbf{c}_1 + \mathbf{c}_2 + \mathbf{c}_3 = \begin{vmatrix} 1 \\ -1 \\ 0 \end{vmatrix} + \begin{vmatrix} 0 \\ 1 \\ -1 \end{vmatrix} + \begin{vmatrix} -1 \\ 0 \\ 1 \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 0 \end{vmatrix}$$

Because non-zero weights $\mathbf{x} = [1, 1, 1]^T$ produce $C\mathbf{x} = \mathbf{0}$, the columns of C are **linearly dependent**! Its columns do not span all of $\mathbb{R}^3$; they span only a 2D plane perpendicular to $[1, 1, 1]^T$. The cyclic difference system $C\mathbf{x} = \mathbf{b}$ is solvable if and only if $b_1 + b_2 + b_3 = 0$.

Part VI: Step-by-Step Worked Problems with Fast-Tracks

Worked Problem 1: 3D Vector Scaling, Norms & Direction Cosine Verification **INTEGER NORM TRIPLES**

1. PROBLEM STATEMENT

Given two vectors in $\mathbb{R}^3$: $\mathbf{u} = [2, -3, 6]^T$ and $\mathbf{v} = [-1, 2, 2]^T$:
(a) Compute the composite linear combination $\mathbf{w} = 3\mathbf{u} - 2\mathbf{v}$.
(b) Calculate the exact Euclidean norm $\|\mathbf{w}\|$.
(c) Find the normalized unit vector $\hat{\mathbf{w}}$ in the direction of $\mathbf{w}$.
(d) Compute the direction cosines ($\cos\alpha$, $\cos\beta$, $\cos\gamma$) relative to the Cartesian coordinate axes.

2. GIVEN PARAMETERS & KNOWNS

$\mathbf{u} = [2, -3, 6]^T$, $\mathbf{v} = [-1, 2, 2]^T$, $c_1 = 3$, $c_2 = -2$.

3. ANALYTICAL & NUMERICAL DERIVATION

Step (a): Compute the linear combination entry-by-entry:

$$\mathbf{w} = 3\begin{vmatrix} 2 \\ -3 \\ 6 \end{vmatrix} - 2\begin{vmatrix} -1 \\ 2 \\ 2 \end{vmatrix} = \begin{vmatrix} 6 \\ -9 \\ 18 \end{vmatrix} - \begin{vmatrix} -2 \\ 4 \\ 4 \end{vmatrix} = \begin{vmatrix} 6-(-2) \\ -9-4 \\ 18-4 \end{vmatrix} = \begin{vmatrix} 8 \\ -13 \\ 14 \end{vmatrix}$$

Step (b): Compute Euclidean norm:

$$\|\mathbf{w}\| = \sqrt{(8^2 + (-13)^2 + 14^2)} = \sqrt{(64 + 169 + 196)} = \sqrt{(429)} \approx 20.7123$$

Step (c): Construct the unit vector:

$$\hat{\mathbf{w}} = \frac{1}{\sqrt{429}}\begin{vmatrix} 8 \\ -13 \\ 14 \end{vmatrix} = \begin{vmatrix} 8/\sqrt{429} \\ -13/\sqrt{429} \\ 14/\sqrt{429} \end{vmatrix}$$

Step (d): Compute direction cosines:

$$\cos\alpha = \hat{\mathbf{w}} \cdot \mathbf{e}_1 = \frac{8}{\sqrt{429}} \approx 0.3862, \quad \cos\beta = \frac{-13}{\sqrt{429}} \approx -0.6276, \quad \cos\gamma = \frac{14}{\sqrt{429}} \approx 0.6759$$

4. FINAL ANSWERS

$\mathbf{w} = [8, -13, 14]^T$, $\|\mathbf{w}\| = \sqrt{429}$, $\hat{\mathbf{w}} = [8, -13, 14]^T / \sqrt{429}$.

Sanity Check: Verify the Pythagorean sum of direction cosines: $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = (64 + 169 + 196) / 429 = 429 / 429 = 1.00000$. Unit magnitude strictly confirmed.

Worked Problem 2: Span Membership & Planar Redundancy in $\mathbb{R}^3$ AUGMENTED ROW REDUCTION

1. PROBLEM STATEMENT

Given two generating vectors $\mathbf{v}_1 = [1, 2, 1]^T$ and $\mathbf{v}_2 = [3, 0, -1]^T$:

(a) Determine whether the target vector $\mathbf{b}_1 = [5, 4, 1]^T$ lies in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$.

(b) Determine whether $\mathbf{b}_2 = [2, 1, 3]^T$ lies in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$.

(c) Confirm geometrically using the normal vector $\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2$.

2. GIVEN PARAMETERS & KNOWNS

$\mathbf{v}_1 = [1, 2, 1]^T$, $\mathbf{v}_2 = [3, 0, -1]^T$, $\mathbf{b}_1 = [5, 4, 1]^T$, $\mathbf{b}_2 = [2, 1, 3]^T$.

3. ANALYTICAL & NUMERICAL DERIVATION

Step (a): Set up the vector equation $c_1\mathbf{v}_1 + c_2\mathbf{v}_2 = \mathbf{b}_1$:

$$\begin{vmatrix} 1 & 3 \\ 2 & 0 \\ 1 & -1 \end{vmatrix} \begin{vmatrix} c_1 \\ c_2 \end{vmatrix} = \begin{vmatrix} 5 \\ 4 \\ 1 \end{vmatrix}$$

From Row 2: $2c_1 + 0 = 4$ &implies; $\mathbf{c_1 = 2}$.

From Row 1: $1(2) + 3c_2 = 5$ &implies; $3c_2 = 3$ &implies; $\mathbf{c_2 = 1}$.

Substitute into Row 3 to verify consistency: $1(2) - 1(1) = 2 - 1 = 1$. Consistent!

Thus, $\mathbf{b}_1 \in$ Span$\{\mathbf{v}_1, \mathbf{v}_2\}$ with unique weights $\mathbf{c} = [2, 1]^T$.

Step (b): Test target vector $\mathbf{b_2} = [2, 1, 3]^T$:

From Row 2: $2c_1 = 1$ &implies; $c_1 = 1/2$.

From Row 1: $1(1/2) + 3c_2 = 2$ &implies; $3c_2 = 3/2$ &implies; $c_2 = 1/2$.

Substitute into Row 3: $1(1/2) - 1(1/2) = 0 \neq 3$. Inconsistency $(0 = 3)$!
No scalar solution exists; therefore, $\mathbf{b}_2 \notin$ Span$\{\mathbf{v}_1, \mathbf{v}_2\}$.

Step (c): Geometrical confirmation via Normal Vector:

$$\mathbf{n} = \mathbf{v}_1 \times \mathbf{v}_2 = \det \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & 1 \\ 3 & 0 & -1 \end{vmatrix} = \mathbf{i}(-2 - 0) - \mathbf{j}(-1 - 3) + \mathbf{k}(0 - 6) = \begin{vmatrix} -2 \\ 4 \\ -6 \end{vmatrix}$$

Check planar dot products:
• $\mathbf{n} \cdot \mathbf{b}_1 = (-2)(5) + (4)(4) + (-6)(1) = -10 + 16 - 6 = \mathbf{0}$. (Strictly lies in plane!)
• $\mathbf{n} \cdot \mathbf{b}_2 = (-2)(2) + (4)(1) + (-6)(3) = -4 + 4 - 18 = \mathbf{-18 \neq 0}$. (Strictly outside plane!)

4. FINAL ANSWERS

$\mathbf{b}_1$ is in the span (weights $c_1 = 2$, $c_2 = 1$). $\mathbf{b}_2$ is not in the span.

Sanity Check: $2\mathbf{v}_1 + 1\mathbf{v}_2 = [2(1)+3, 2(2)+0, 2(1)-1]^T = [5, 4, 1]^T = \mathbf{b}_1$. Exact match.

Worked Problem 3: Orthogonal Decomposition & Minimum Distance to a Line PYTHAGOREAN SPLIT

1. PROBLEM STATEMENT

Given point vector $\mathbf{v} = [4, 1, 2]^T$ and line direction vector $\mathbf{d} = [2, -2, 1]^T$ in $\mathbb{R}^3$:
(a) Find the orthogonal projection vector $\mathbf{p} = \text{proj}_{\mathbf{d}}(\mathbf{v})$.

(b) Calculate the orthogonal error vector $\mathbf{e} = \mathbf{v} - \mathbf{p}$.

(c) Determine the exact shortest Euclidean distance from $\mathbf{v}$ to the line spanned by $\mathbf{d}$.

(d) Verify the Pythagorean theorem $\|\mathbf{v}\|^2 = \|\mathbf{p}\|^2 + \|\mathbf{e}\|^2$.

2. GIVEN PARAMETERS & KNOWNS

$\mathbf{v} = [4, 1, 2]^T$, $\mathbf{d} = [2, -2, 1]^T$.

3. ANALYTICAL & NUMERICAL DERIVATION

Step (a): Compute the projection scalar $c = (\mathbf{v} \cdot \mathbf{d}) / (\mathbf{d} \cdot \mathbf{d})$:

$$\mathbf{v} \cdot \mathbf{d} = (4)(2) + (1)(-2) + (2)(1) = 8 - 2 + 2 = 8$$

$$\mathbf{d} \cdot \mathbf{d} = 2^2 + (-2)^2 + 1^2 = 4 + 4 + 1 = 9$$

$$c = 8 / 9 \quad \&\text{implies;} \quad \mathbf{p} = \frac{8}{9} \begin{vmatrix} 2 \\ -2 \\ 1 \end{vmatrix} = \begin{vmatrix} 16/9 \\ -16/9 \\ 8/9 \end{vmatrix}$$

Step (b): Compute the orthogonal error vector $\mathbf{e} = \mathbf{v} - \mathbf{p}$:

$$\mathbf{e} = \begin{vmatrix} 4 \\ 1 \\ 2 \end{vmatrix} - \begin{vmatrix} 16/9 \\ -16/9 \\ 8/9 \end{vmatrix} = \begin{vmatrix} 36/9 - 16/9 \\ 9/9 - (-16/9) \\ 18/9 - 8/9 \end{vmatrix} = \frac{1}{9} \begin{vmatrix} 20 \\ 25 \\ 10 \end{vmatrix} = \begin{vmatrix} 20/9 \\ 25/9 \\ 10/9 \end{vmatrix}$$

Step (c): Minimum distance = $\|\mathbf{e}\|$:

$$\|\mathbf{e}\| = \frac{1}{9} \sqrt{(20^2 + 25^2 + 10^2)} = \frac{1}{9} \sqrt{(400 + 625 + 100)} = \frac{\sqrt{1125}}{9} = \frac{15\sqrt{5}}{9} = \frac{5\sqrt{5}}{3} \approx 3.7268$$

Step (d): Orthogonality & Pythagorean verification:

$$\mathbf{e} \cdot \mathbf{d} = \frac{1}{9} [(20)(2) + (25)(-2) + (10)(1)] = \frac{1}{9} [40 - 50 + 10] = 0 / 9 = 0. \quad (\perp \text{ verified!})$$

$$\|\mathbf{p}\|^2 = (8/9)^2 \|\mathbf{d}\|^2 = (64/81) \times 9 = 64/9.$$

$$\|\mathbf{e}\|^2 = 1125 / 81 = 125 / 9.$$

$$\|\mathbf{p}\|^2 + \|\mathbf{e}\|^2 = 64/9 + 125/9 = 189/9 = 21.$$

$$\|\mathbf{v}\|^2 = 4^2 + 1^2 + 2^2 = 16 + 1 + 4 = 21. \quad \text{(Exact Pythagorean match!)}$$

4. FINAL ANSWERS

$\mathbf{p} = [16/9, -16/9, 8/9]^T$, $\mathbf{e} = [20/9, 25/9, 10/9]^T$, Shortest Distance = $5\sqrt{5}/3$.

Sanity Check: $\|\mathbf{p}\|^2 + \|\mathbf{e}\|^2 = 21 = \|\mathbf{v}\|^2$. The projection lies closer to origin than $\mathbf{v}$ ($\|\mathbf{p}\| \leq \|\mathbf{v}\|$), strictly satisfying Cauchy-Schwarz.

Worked Problem 4: Cauchy-Schwarz & Triangle Inequality Tightness in $\mathbb{R}^4$ — 4D VECTOR GEOMETRY

1. PROBLEM STATEMENT

Given two vectors in 4D space $\mathbb{R}^4$: $\mathbf{x} = [1, 2, 2, 4]^T$ and $\mathbf{y} = [2, 1, -2, 1]^T$:
(a) Compute the inner product $\mathbf{x} \cdot \mathbf{y}$ and the norms $\|\mathbf{x}\|$, $\|\mathbf{y}\|$.
(b) Calculate the exact angle θ between $\mathbf{x}$ and $\mathbf{y}$ in degrees.
(c) Evaluate the Cauchy-Schwarz ratio $|\mathbf{x} \cdot \mathbf{y}| \,/\, (\|\mathbf{x}\| \, \|\mathbf{y}\|)$.
(d) Verify the Triangle Inequality $\|\mathbf{x} + \mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$ and compute the slack gap.

2. GIVEN PARAMETERS & KNOWNS

$\mathbf{x} = [1, 2, 2, 4]^T$, $\mathbf{y} = [2, 1, -2, 1]^T$.

3. ANALYTICAL & NUMERICAL DERIVATION

Step (a): Compute dot product and norms:

$$\mathbf{x} \cdot \mathbf{y} = (1)(2) + (2)(1) + (2)(-2) + (4)(1) = 2 + 2 - 4 + 4 = 4$$

$$\|\mathbf{x}\| = \sqrt{(\,1^2 + 2^2 + 2^2 + 4^2\,)} = \sqrt{(\,1 + 4 + 4 + 16\,)} = \sqrt{25} = 5$$

$$\|\mathbf{y}\| = \sqrt{(\,2^2 + 1^2 + (-2)^2 + 1^2\,)} = \sqrt{(\,4 + 1 + 4 + 1\,)} = \sqrt{10} \approx 3.1623$$

Step (b): Compute angle θ:

$$\cos\theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \, \|\mathbf{y}\|} = \frac{4}{5\sqrt{10}} = \frac{4\sqrt{10}}{50} = \frac{2\sqrt{10}}{25} \approx 0.25298$$

$$\theta = \arccos(0.25298) \approx 75.35^\circ \; (1.315 \text{ rad})$$

Step (c): Evaluate Cauchy-Schwarz bound:

$$|\,\mathbf{x} \cdot \mathbf{y}\,| = 4, \qquad \|\mathbf{x}\| \, \|\mathbf{y}\| = 5\sqrt{10} \approx 15.8114$$

Ratio = 4 / 15.8114 = 0.2530 ≤ 1.0000. (Strict Cauchy-Schwarz inequality verified!)

Step (d): Triangle Inequality evaluation:

$$\mathbf{x} + \mathbf{y} = \begin{vmatrix} 1+2 \\ 2+1 \\ 2-2 \\ 4+1 \end{vmatrix} = \begin{vmatrix} 3 \\ 3 \\ 0 \\ 5 \end{vmatrix}$$

$$\|\mathbf{x} + \mathbf{y}\| = \sqrt{(\,3^2 + 3^2 + 0^2 + 5^2\,)} = \sqrt{(\,9 + 9 + 0 + 25\,)} = \sqrt{43} \approx 6.5574$$

$$\|\mathbf{x}\| + \|\mathbf{y}\| = 5 + \sqrt{10} \approx 5 + 3.1623 = 8.1623$$

Gap $\delta = (\|\mathbf{x}\| + \|\mathbf{y}\|) - \|\mathbf{x} + \mathbf{y}\| = 8.1623 - 6.5574 = 1.6049 > 0$.

4. FINAL ANSWERS

$\mathbf{x} \cdot \mathbf{y} = 4$, $\|\mathbf{x}\| = 5$, $\|\mathbf{y}\| = \sqrt{10}$, $\theta \approx 75.35^\circ$, $\|\mathbf{x}+\mathbf{y}\| = \sqrt{43} \le 5+\sqrt{10}$.

Sanity Check: Since $\theta = 75.35° \neq 0°$, the vectors are non-collinear; therefore, the Cauchy-Schwarz ratio is strictly < 1 and the triangle inequality gap is strictly > 0.

Worked Problem 5: Column Picture of a 3×3 Difference System Ax = b **STRANG'S COLUMN SYNTHESIS**

1. PROBLEM STATEMENT

Given the 3 × 3 first-difference matrix A and target output vector **b**:

$$A = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{vmatrix}, \quad \mathbf{b} = \begin{vmatrix} 3 \\ 2 \\ 4 \end{vmatrix}$$

(a) Express A**x** = **b** explicitly as a linear combination of the columns $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$ of A.

(b) Solve for the combination multipliers $\mathbf{x} = [x_1, x_2, x_3]^T$ using forward column accumulation.

(c) Show that **x** = S**b**, where S is the lower-triangular summation matrix.

(d) Verify by direct column vector substitution.

2. GIVEN PARAMETERS & KNOWNS

$\mathbf{a}_1 = [1, -1, 0]^T$, $\mathbf{a}_2 = [0, 1, -1]^T$, $\mathbf{a}_3 = [0, 0, 1]^T$, $\mathbf{b} = [3, 2, 4]^T$.

3. ANALYTICAL & NUMERICAL DERIVATION

Step (a): Write the column combination form:

$$x_1 \begin{vmatrix} 1 \\ -1 \\ 0 \end{vmatrix} + x_2 \begin{vmatrix} 0 \\ 1 \\ -1 \end{vmatrix} + x_3 \begin{vmatrix} 0 \\ 0 \\ 1 \end{vmatrix} = \begin{vmatrix} 3 \\ 2 \\ 4 \end{vmatrix}$$

Step (b): Solve by forward accumulation:

- Row 1: $x_1 = 3$.
- Row 2: $-x_1 + x_2 = 2$ &implies; $x_2 = x_1 + 2 = 3 + 2 = \mathbf{5}$.
- Row 3: $-x_2 + x_3 = 4$ &implies; $x_3 = x_2 + 4 = 5 + 4 = \mathbf{9}$.

Thus, the weight vector is $\mathbf{x} = [3, 5, 9]^T$.

Step (c): Connect to the Summation Matrix $S = A^{-1}$:

$$\mathbf{x} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} \begin{vmatrix} 3 \\ 2 \\ 4 \end{vmatrix} = \begin{vmatrix} 3 \\ 3+2 \\ 3+2+4 \end{vmatrix} = \begin{vmatrix} 3 \\ 5 \\ 9 \end{vmatrix}$$

The multipliers x_k compute the running cumulative sum (discrete integral) of the target values b_k!

Step (d): Verify by direct column synthesis:

$$3\begin{bmatrix}1\\-1\\0\end{bmatrix}+5\begin{bmatrix}0\\1\\-1\end{bmatrix}+9\begin{bmatrix}0\\0\\1\end{bmatrix}=\begin{bmatrix}3\\-3\\0\end{bmatrix}+\begin{bmatrix}0\\5\\-5\end{bmatrix}+\begin{bmatrix}0\\0\\9\end{bmatrix}=\begin{bmatrix}3+0+0\\-3+5+0\\0-5+9\end{bmatrix}=\begin{bmatrix}3\\2\\4\end{bmatrix}=\mathbf{b}$$

4. FINAL ANSWERS

Weight vector $\mathbf{x} = [3, 5, 9]^T$. Difference matrix inversion produces running cumulative integration.

Sanity Check: $A \cdot \mathbf{x} = [3, 5-3, 9-5]^T = [3, 2, 4]^T = \mathbf{b}$. Exact recovery confirmed.

Part VII: Master Reference Appendices

Appendix A: Comprehensive Vector & Dot Product Formula Sheet

Concept / Quantity	Standard Formula	Key Notes & Conditions
Dot Product (Inner Product)	$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T\mathbf{v} = \sum_{i=1}^{n} u_i v_i = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta$	Commutative, bilinear, positive-definite
Euclidean Norm (L2 Length)	$\|\mathbf{v}\| = \sqrt{(\mathbf{v} \cdot \mathbf{v})} = \sqrt{\left(\sum_{i=1}^{n} v_i^2 \right)}$	$\|\mathbf{v}\| \geq 0$; $\|c\mathbf{v}\| = \|c\| \, \|\mathbf{v}\|$
Unit Direction Vector	$\hat{\mathbf{u}} = \mathbf{v} / \|\mathbf{v}\|$	$\|\hat{\mathbf{u}}\| = 1$ (defined for $\mathbf{v} \neq \mathbf{0}$)
Orthogonality Condition	$\mathbf{u} \perp \mathbf{v} \quad \Leftrightarrow \quad \mathbf{u} \cdot \mathbf{v} = 0$	$\cos(90°) = 0$; Pythagorean theorem holds
Scalar Projection	$\text{comp}_{\mathbf{w}}(\mathbf{v}) = \|\mathbf{v}\| \cos\theta = (\mathbf{v} \cdot \mathbf{w}) / \|\mathbf{w}\|$	Signed scalar length along $\mathbf{w}$
Vector Projection	$\text{proj}_{\mathbf{w}}(\mathbf{v}) = [(\mathbf{v} \cdot \mathbf{w}) / (\mathbf{w} \cdot \mathbf{w})] \, \mathbf{w}$	Decomposes $\mathbf{v} = \mathbf{p} + \mathbf{e}$ with $\mathbf{e} \perp \mathbf{w}$
Cauchy-Schwarz Inequality	$\| \mathbf{u} \cdot \mathbf{v} \| \leq \|\mathbf{u}\| \, \|\mathbf{v}\|$	Equality $\Leftrightarrow$ $\mathbf{u}$ and $\mathbf{v}$ are collinear
Triangle Inequality	$\| \mathbf{u} + \mathbf{v} \| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$	Equality $\Leftrightarrow$ $\mathbf{u}$ and $\mathbf{v}$ point in same direction
Matrix-Vector Product	$A\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{a}_j$	Column picture: combination of columns

Appendix B: Inspection Rules & Mental Calculation Cheat Sheet

Rule 1: Instant Dot Product Signs

Check the dominant coordinate signs: if vectors have identical signs (+, +) or (−, −), their product is strictly positive ($\theta < 90°$). If signs oppose (+, −), check if positive products dominate or if total sums to zero.

Rule 2: 2D Orthogonal Rotation

To rotate $[a, b]^T$ by 90° counterclockwise, take $[-b, a]^T$. To rotate clockwise, take $[b, -a]^T$. Never solve a linear system for a 2D normal vector!

Rule 3: Integer Norm Recognition

Whenever you see coordinates [1, 2, 2], the norm is immediately **3**. For [2, 3, 6], the norm is **7**. For [1, 4, 8], the norm is **9**. For [2, 6, 9], the norm is **11**. Factoring out common factors makes square roots virtually disappear.

Rule 4: Target Vector Span Inspection

In A$\mathbf{x}$ = $\mathbf{b}$, before Gaussian elimination: if $\mathbf{b}$ is the first column, $\mathbf{x}$ = $[1, 0, 0]^T$; if $\mathbf{b}$ is the sum of columns, $\mathbf{x}$ = $[1, 1, 1]^T$; if sum of rows of A is zero while sum of $\mathbf{b}$ is non-zero, the system is immediately inconsistent (no solution).